D03 · Maintenance & Asset Reliability

05 · MTBF & reliability KPI framework

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1 WHAT IT IS

The reliability KPI framework is the disciplined set of metrics that diagnose equipment reliability — MTBF (mean time between failures), MTTR (mean time to repair), availability, and the underlying failure-mode decomposition that converts the aggregate metrics into actionable diagnoses. The Weibull analysis underlying the framework characterizes whether failures are infant-mortality (beta < 1), random (beta ~ 1), or wear-out (beta > 1), with each pattern requiring a fundamentally different maintenance strategy. Aggregate metrics without decomposition are summary statistics; decomposed metrics with Weibull characterization are the diagnostic apparatus that drives reliability improvement.

2 WHY IT MATTERS

The reliability KPI framework is the most universally reported and most consistently mis-used set of operational metrics in industrial maintenance. **The dominant failure mode is MTBF reported as a single aggregate number — "MTBF is 480 hours" — without decomposition by failure mode, by asset criticality, or by Weibull shape parameter. The aggregate hides three different reliability problems with three different strategic remedies, and the maintenance organization attacks the average instead of the underlying patterns**. The international standard for reliability data collection in industrial operations is explicit on the decomposition requirement **(ISO 14224:2016 — international standard, "Petroleum, petrochemical and natural gas industries — Collection and exchange of reliability and maintenance data for equipment")**: reliability data must be captured and reported with failure-mode classification, equipment-level granularity, and time-to-failure distributions that support Weibull analysis. Industry data confirms the asymmetry — **operations that decompose MTBF by failure mode and characterize failures with Weibull analysis typically deliver 30–60% improvement in critical-asset MTBF within 18 months; operations that report aggregate MTBF without decomposition show no measurable improvement despite the same maintenance investment**. The difference is not effort; it is whether the metrics support diagnosis.

3 HOW IT IS DONE

1. **Capture failure data at the failure-mode level — not at the asset-failed level.** A pump that fails because of a bearing problem, a pump that fails because of a seal problem, and a pump that fails because of an impeller problem are three different events with three different causes and three different countermeasures. Recording all three as "pump failure" destroys the diagnostic information. The CMMS work-order completion has to capture failure mode using a standardized taxonomy (ISO 14224 provides one for process industries; equivalent standards exist for other industries) so that aggregate metrics can be decomposed downstream.
2. **Run Weibull analysis on the failure data — and let the shape parameter drive the maintenance strategy.** Weibull shape parameter (beta) less than 1: infant-mortality failures, the strategy is better installation/commissioning and stricter QC at startup. beta approximately 1: random failures, the strategy is condition-based monitoring and run-to-failure where consequence permits (scheduled replacement does not help because failure timing is unpredictable). beta greater than 1: wear-out failures, the strategy is scheduled replacement before the wear-out distribution kicks in. Plants that apply scheduled replacement to random failures waste resources; plants that wait for wear-out failures to manifest catch them at the wrong end of the cost curve.
3. **Decompose MTBF by asset criticality — the average across all assets is meaningless if it hides how critical assets are performing.** An operation with 340 assets at average MTBF of 480 hours might have 22 critical assets at 220 hours MTBF and 318 non-critical assets at 510 hours MTBF — the average looks acceptable while the critical positions are failing every 9 days. Reporting MTBF by criticality tier (critical / important / standard) exposes where the reliability problem actually lives; reporting aggregate MTBF obscures it.
4. **Report MTBF, MTTR, and availability together — none of them alone tells the story.** MTBF measures failure frequency, MTTR measures repair time, availability = MTBF / (MTBF + MTTR) integrates both. A plant with MTBF 600 hours and MTTR 60 hours has 91% availability; a plant with MTBF 800 hours and MTTR 200 hours has 80% availability despite the better

MTBF. Improving MTBF while ignoring MTTR is the most common organizational dysfunction in reliability programs because MTBF improvement is what reliability engineers focus on and MTTR is what maintenance technicians own — the two functions optimize separately and produce sub-optimal availability.

5. **Use survival analysis and reliability growth tracking — not just point-in-time MTBF.** A plant's MTBF this quarter compared to last quarter is a comparison of two estimates each based on small samples of failures; the noise frequently exceeds the signal. The right view is survival curves over time (showing how the failure-time distribution is shifting), reliability growth tracking on specific failure modes after countermeasures, and confidence intervals around MTBF estimates. Plants that report MTBF as a single number with no uncertainty estimate routinely chase noise; plants that report distributions with confidence intervals make decisions on real signal.
6. **Govern the reliability KPI framework jointly between reliability engineering and operations — the metrics belong to both functions or to neither.** The most common organizational pathology is reliability engineering owning the metrics and operations not consulting them in production decisions. The structural fix is monthly reliability review with operations leadership where the failure-mode decomposition, the Weibull characterization, and the action items get reviewed — not as a maintenance report but as an operational diagnostic. When operations leadership engages with the KPI framework, the framework drives decisions; when only reliability engineering looks at the data, it produces reports that nobody acts on.

WORKED EXAMPLE — ILLUSTRATIVE

A chemicals manufacturer (1 plant, 340 production assets including reactors, distillation columns, heat exchangers, pumps, compressors) reports plant-level **aggregate MTBF = 480 hours** monthly. The reliability program has been "improving MTBF" for three years with limited results. Weibull decomposition by failure mode and asset reveals the structure the aggregate was hiding: **73% of total failures concentrate in 22 critical assets (out of 340); within those 22 critical assets, failure modes distribute as 40% wear-out (beta = 3.2 — clear wear-out signature, scheduled replacement is the right strategy), 31% random (beta = 1.0 — condition-based monitoring is the right strategy), 29% infant-mortality (beta = 0.6 — installation and commissioning quality is the issue)**. The aggregate MTBF of 480 hours was simultaneously the average of failure rates that needed three completely different strategies. The implementation: **(1) scheduled replacement program on the top 8 wear-out items (bearings, seals, gaskets on critical positions) before the wear-out distribution kicks in; (2) installation-procedure standardization with stricter QC at commissioning for the 29% of failures showing infant-mortality signature; (3) condition-based monitoring deployed on the random-failure items where prediction is feasible; (4) monthly reliability review with operations leadership where decomposed metrics drive specific action items**. Results 16 months later: **aggregate MTBF 480 -> 720 hours (+50%), but more importantly: unplanned failures on critical assets reduced 71%, downtime cost reduced 58%, MTBF improvement concentrated where the operation actually needed it**. The aggregate number improvement is real but secondary; the structural improvement came from the decomposition that the aggregate had been hiding.

IN PRACTICE

Decompose MTBF by failure mode using ISO 14224 taxonomy

Aggregate MTBF hides three different reliability problems with three different strategic remedies. The standardized failure-mode taxonomy **(ISO 14224:2016 — international standard)** is the data-collection structure that makes decomposition possible; without it the data exists but cannot be analyzed.

Let the Weibull shape parameter drive the maintenance strategy

beta < 1 (infant mortality) -> installation quality is the problem, not maintenance frequency. beta ~ 1 (random) -> condition monitoring is appropriate; time-based maintenance wastes effort. beta > 1 (wear-out) -> scheduled replacement before the wear-out zone. Same MTBF, three different strategies.

Decompose by asset criticality — critical-asset performance hides in the average

A plant-wide MTBF of 480 hours can mean every asset performs identically or that critical assets fail every 96 hours while non-critical assets last 2,400. Same average, radically different operations. Critical-asset MTBF is the metric that drives intervention; aggregate MTBF is the metric that goes in the report.

Monthly joint review: reliability engineering + operations management

Quarterly Weibull analyses produced by reliability engineering that operations does not read; monthly maintenance decisions made by operations that reliability does not influence — the metric framework dies in the gap. Monthly joint review with explicit translation from failure-mode/Weibull findings to budget and strategy decisions is the structural fix.

WHAT YOU WOULD USE

RELIABILITY KPI & WEIBULL ANALYSIS

ReliaSoft Weibull++ / Synthesis Platform (Hottinger) / Reliasoft RGA — Dedicated reliability engineering platforms with native Weibull analysis, reliability growth tracking, survival analysis, FRACAS integration, and ISO 14224-aligned data structures; the standard tool set for reliability engineering functions in process and discrete manufacturing **(Hottinger Brüel & Kjær — vendor)**.

IBM Maximo Reliability / GE Vernova APM / AVEVA Asset Performance Management — Enterprise asset performance management platforms that integrate reliability analytics directly with the CMMS/EAM operational data, producing decomposed KPIs as a native output rather than a separate analytical artifact **(IBM — vendor; GE Vernova — vendor; AVEVA — vendor)**.

Minitab / JMP (SAS) with Weibull modules + structured CMMS data extraction — General-purpose statistical platforms with strong reliability-analysis capability, paired with structured data extraction from the operation's CMMS; the right fit when reliability engineering has statistical depth in-house and the platform purchase decision goes to general statistical tools **(Minitab — vendor; SAS Institute — vendor)**.

R or Python (with reliability/survival packages) + structured CMMS export + monthly joint review — For operations with data-engineering capability — open-source statistical environments (R with the survival package, Python with lifelines and SciPy), structured monthly export from CMMS into the analytical environment, joint reliability + operations review of the decomposed outputs. The lowest-tech option produces the deepest analytical sophistication when the in-house capability is real.

4 SO WHAT

The MTBF reported as a single aggregate number, without Weibull decomposition by failure mode and criticality, is an average that does not diagnose — and averages do not improve, diagnoses do. Capture failure data at the failure-mode level using a standardized taxonomy, run Weibull analysis and let the shape parameter drive the maintenance strategy, decompose MTBF by asset criticality so critical-asset performance is not hidden in the average, report MTBF + MTTR + availability together, use survival analysis and reliability growth tracking instead of point-in-time MTBF, and govern the framework jointly between reliability engineering and operations. The same data set produces no improvement when reported as aggregate MTBF and 30–60% critical-asset MTBF improvement when reported as decomposed metrics with Weibull characterization — the difference is structural, and it lives in whether the metrics support diagnosis or just summary.

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